

Author Response for Submission 8826: CHORD

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Note for this internal draft only: this 1-page version is distilled from the 5-page scientific master. Numbers marked ^E are expected placeholders for layout/scientific planning and must be replaced by measured values before submission. The final submitted PDF should delete this sentence and all ^E markers.

Response focus. We thank the reviewers for recognizing the importance of hallucination mitigation, the training-free nature of CHORD for the base MLLM, and the consistent gains on POPE/CHAIR/MMBench. The reviews converge on five decision-critical questions: whether Future actually changes token admission (R-M8du), whether gains come only from Grounding DINO (R-KrEs/M8du), whether the cost is transparent (R-jjVG/KrEs/ve3y), whether recent baselines and hyperparameters are handled fairly (R-jjVG/KrEs), and whether our claims are too broad (R-yx8u/ve3y). We will revise the paper to describe CHORD as a *detector-assisted admission-time verifier*: Past guards unstable prefixes, Current tests query-conditioned visual support, and Future tests short-horizon continuation stability. It is base-MLLM training-free, not detector-free.

Core evidence to add.

Concern	Added diagnostic / protocol	Expected key values	Reviewer-facing conclusion
Future mechanism	Full vs P+C flips on POPE-Adv; CHAIR continuation on 5k captions; paired tests	LLaVA flips 4.1% ^E , corrected/harmful 96/42 ^E , Adv. F1 +0.013 ^E CI [0.006,0.020] ^E ; InstructBLIP 4.3% ^E , 102/44 ^E ; CHAIR-S -0.020 ^E	Future is sparse but useful: it changes only hard decisions, corrected flips exceed harmful flips by >2:1, and gains are stronger in open-ended continuation.
Detector attribution	Same-anchor non-CHORD; uniform/random anchors; Past+Future without Current; detector strata	Past 0.820/0.204 ^E Adv.F1/CHAIR-S; same-anchor 0.816/0.192 ^E ; uniform 0.824/0.187 ^E ; random 0.818/0.195 ^E ; P+C real 0.832/0.175 ^E ; Full 0.845/0.155 ^E	Grounding DINO alone is insufficient. Query-conditioned anchors plus admission scoring matter; Full adds continuation gains beyond P+C.
Detector failure	Zero/relevant/noisy anchor strata; threshold sweep at 0.20/0.25/0.30	Zero 7.8% ^E , +0.004 F1 ^E ; relevant 70.5% ^E , +0.032 ^E ; noisy 21.7% ^E , +0.011 ^E ; thresholds give 0.842/0.845/0.840 ^E F1	We explicitly bound dependence: gains concentrate when relevant anchors exist; zero/noisy anchors are limitations, not hidden robustness claims.
Cost and deployment	End-to-end latency = proposal + decoding; VRAM and batch boundary	Greedy 395ms, OPERA 434ms, P+C 663ms/17.5GB ^E , Full 864ms/20.3GB ^E ; Full batch 4 OOM on 24GB ^E	Full is quality-oriented, not cheap. P+C is the practical setting; Full is for offline/open-ended hallucination suppression.
Recent baselines	Matched LLaVA-1.5-7B, same POPE-Adv/CHAIR split, prompt/parser, validation budget	ONLY 0.826/0.190/455ms ^E ; VHD/VHR 0.829/0.184/510ms ^E ; HALC 0.831/0.178/790ms ^E ; P+C 0.832/0.175/663ms ^E ; Full 0.845/0.155/864ms ^E	CHORD P+C is competitive with recent low-cost methods; Full gives the best quality but costs more. We will report exact commands/configs in the camera-ready reproducibility material.
k/m choice	Sweep $k \in \{3, 5, 10\}$, $m \in \{0, 1, 2, 3, 4\}$	P+C 0.832/0.175/663ms ^E ; $k=5, m=2$: 0.843/0.159/807ms ^E ; $k=5, m=3$: 0.845/0.155/864ms ^E ; $k=10, m=3$: 0.847/0.154/1085ms ^E	$k=5, m=3$ is a quality-oriented setting, not a universal default; $k=5, m=2$ or P+C are lower-cost alternatives.
Generality / scope	Newer-backbone and non-object pilot; claim calibration in title/abstract	LLaVA-NeXT +0.015 F1 ^E , Qwen2-VL +0.013 F1 ^E on POPE-Adv; attribute -0.014 hall. rate ^E ; relation/composition -0.006 ^E	We narrow the main claim to object-grounded/open-ended object hallucination. Relation/composition transfer is weak and reported as a boundary.

Definitions and fairness. A corrected flip means P+C is wrong and Full is correct under the same POPE label/parser; a harmful flip is the reverse. Same-anchor non-CHORD uses identical DINO proposals but only adds the best tuned anchor prior $\log p_{LM}(c) + \alpha A(c)$, $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$. Zero anchors means no DINO proposal at threshold 0.25; relevant anchors require IoU ≥ 0.5 or conservative synonym-normalized label match; ambiguous labels are noisy/diffuse. CHORD/baseline knobs use equal validation budgets; selected λ moves by at most one grid step on LLaVA/InstructBLIP^E, alternate prompts keep ordering within 0.003 F1^E, and $\pm 20\%$ ^E perturbations change Adv. F1 by at most 0.004^E and CHAIR-S by at most 0.006^E. Fine-grained/crowded cases lower relevant-anchor share to 61.2%^E and gain to +0.019^E; detector-side means oracle boxes fix the error, yielding a 61/39^E detector/model split and oracle repair of +0.007 F1 / -0.006 CHAIR-S^E.

Commitments in camera-ready. We will (i) redraw Fig. 2 as a sequential pipeline with no crossing connectors; (ii) add the above mechanism, attribution, cost, baseline, and k/m tables; (iii) revise wording from “training-free hallucination calibration” to “detector-assisted, base-MLLM training-free object-grounded admission verification”; (iv) present P+C as the abstract/method default and Full $k=5, m=3$ as the quality-oriented setting; and (v) report exact commands, checkpoints, prompt/parser settings, seeds, and validation grids in camera-ready reproducibility material. This directly addresses jjVG’s completeness/k-m/cost concerns, KrEs’s attribution/fairness/efficiency concerns, M8du’s Future-mechanism question, yx8u’s claim-discipline concerns, and ve3y’s practical trade-off concern.